

need to operate them. For example, your three CFL lights (16W×3), a fan (40W) and a TV (60W) will add up to 150W. This is the minimum power rating that you need for your inverter. It is necessary to buy an inverter with at least 50 per cent or even 100 per cent more capacity in watts.

However, for finding your energy needs, you need to multiply the watt rating of the appliance by the time in hours you need to use it. For instance, in the above example, if you need to run these appliances for five hours, your energy need is $150\text{W} \times 5\text{h} = 750\text{ Wh}$. So your inverter could be of 225-300W (continuous) rating but your energy need is higher.

It is important to remember that your battery capacity depends on your energy need. The battery has to be large enough to accumulate from the sun and store 750 Wh of energy and not 150W power as some people often misunderstand.

Your home inverter generally needs 12V or 24V DC input depending on what you buy. This is because lead-acid batteries are often rated at 12V. For 24V, you need to connect two of them in series.

Another rating of the battery is its capacity specified in Ah. A 40Ah battery means that it can deliver 8A for five hours, 2A for 20 hours, etc. In other words, its energy storage capacity is $12\text{V} \times 40\text{ Ah} = 480\text{ Wh}$.

So if you need 750 Wh as in the example given above, you will need a 12V, 80Ah battery that can store $12 \times 80 = 960\text{ Wh}$ energy for night use. Even here, it is essential to buy a battery with 50 to 60 per cent more capacity in order to ensure a longer battery life. Selection criteria is similar to that for a water tank capacity; buy always more than what you need.

So for the most economic size of the solar power system you need, calculate how many hours you may want to

use each of your appliances and find energy need of each by multiplying by its wattage rating. For example, using a 15W CFL light for four hours will require 60 Wh of energy. A 40W fan that you may use all night long—say, for ten hours—will require 400 Wh of energy. Add up energy needs of all the appliances to calculate your energy need in Wh.

Your PV solar modules are rated in peak watts. But, in practice, these, on an average, deliver only 70 to 75 per cent power (over about five hours) of the solar energy that you get to accumulate from the sun. Therefore a panel rated at, say, 100Wp, will give you 70W for five hours—that is, energy of 350 Wh ($5 \times 0.7 \times 100$). So if your energy need is 700 Wh, your PV module should be rated at 200Wp. ●

The author is ex-chairman, Electronics Commission, government of India, and a former advisor to late Rajiv Gandhi

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